Mathematical Analysis

Yurii Belov[1]

[1]St. Petersburg State University

21 октября 2020 г.

Examples

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \ldots + \frac{x^n}{n!} + o(x^n), \quad x \to 0;$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \ldots + \frac{(-1)^n x^{2n-1}}{(2n-1)!} + o(x^{2n-1}), \quad x \to 0;$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \ldots + \frac{(-1)^n x^{2n}}{(2n)!} + o(x^{2n}), \quad x \to 0;$$

Examples

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - ... + (-1)^{n+1}\frac{x^n}{n} + o(x^n);$$

$$(1+x)^\alpha = 1+\alpha x+\frac{\alpha(\alpha-1)}{2!}x^2+....+\frac{\alpha(\alpha-1)...(\alpha-(n-1))x^n}{n!}+o(x^n);$$

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - ... + \frac{(-1)^n x^{2n+1}}{2n+1} + o(x^{2n+1});$$

$$\arcsin x = \sum_{k=1}^{n} \frac{(2k)!}{4^k(k!)^2(2k+1)}x^{2k+1} + o(x^{2k+1}).$$

Catalan numbers and Fibonacci numbers

$$c_0 = 1, \quad c_{n+1} = \sum_{k=0}^{n} c_0 c_{n-k}.$$

$$\sum_{n\geq 0} c_n x^n = ?$$

Balanced bracket sequences and so on...

$$\frac{1}{1 - x - x^2} = \sum_{n\geq 0} F_n x^n.$$

Pointwise convergence and uniform convergence

Definition

We will say that $f_n \to f$ on E if $f_n(x) \to f(x)$ for any $x \in E$.

Definition

We will say that $f_n \to f$ uniformly on E if for any $\varepsilon > 0$ there exists N such that $|f_n(x) - f(x)| < \varepsilon$ for any $n > N$ and $x \in E$.

Stokes-Zaidel theorem

Theorem

Let f_n be a sequence of continuous functions and $f_n \to f$ uniformly on E. Then f is a continuous function.

Theorem (Cauchy)

TFAE

1. $f_n \to f$ uniformly on E;
2. For any $\varepsilon > 0$ there exists N such that for any $k, l > N$ we have $\sup_{x \in E} |f_k(x) - f_l(x)| < \varepsilon$.

Theorem

Let f_n be a sequence of continuous functions and $f_n \to f$ uniformly on E. Then f is a continuous function.

Theorem (Cauchy)

TFAE

1. $f_n \to f$ *uniformly on* E;
2. *For any* $\varepsilon > 0$ *there exists* N *such that for any* $k, l > N$ *we have* $\sup_{x \in E} |f_k(x) - f_l(x)| < \varepsilon$.

Weierstrass theorem

Theorem

Let $u_n : X \mapsto \mathbb{R}$. *If* $|u_n| \le d_n$ *and* $\sum_n d_n < \infty$, *then* $\sum_{n=1}^{\infty} u_n(x)$ *is uniformly convergent series.*

Metric spaces

Definition

We will say that $\{X, d\}$ is a metric space if

1. $$d(x, y) \geq 0, \quad d(x, y) = 0 \Leftrightarrow x = y;$$
2. $$d(x, y) = d(y, x), x, y \in X;$$
3. $$d(x, y) + d(y, z) \geq d(x, z).$$

Theorem

Let $f_n \to f$ on E and f_n be a smooth functions. If f_n' uniformly converges to g, then g is also smooth and $g' = f$.